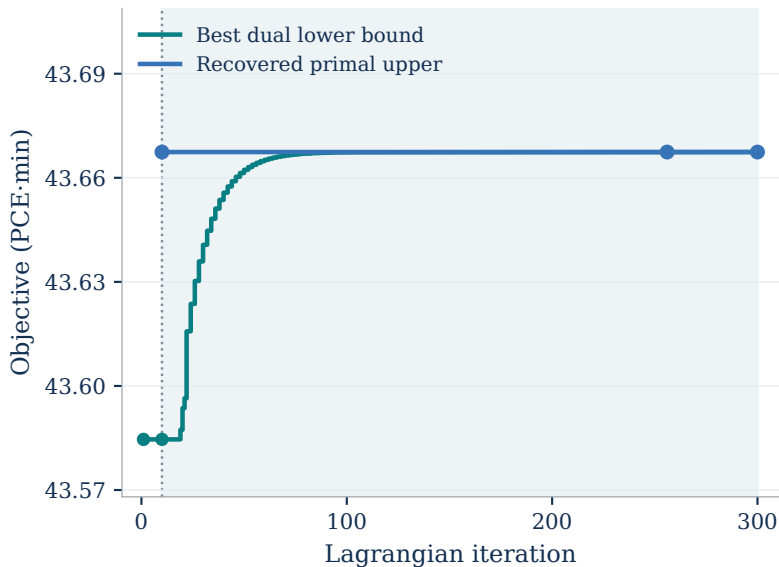


Lagrangian bounds and certified gap

a Best saved bounds**b Certified relative gap**